

↳ DFS / Graph Traversal

Monotonic Stack

- ⇒ A stack that you force to stay sorted, either always increasing or decreasing from bottom to top.
- ⇒ You pop off anything that breaks the order before you push something new.
- ⇒ Once something gets popped, it's gone. Permanently.
- ⇒ Why throw data away permanently?
 - ↳ When you pop something, you've learned the answer about it. Pop is the payoff.
- "For each element, find the nearest other element that is bigger/smaller than it".
- ⇒ How it works:
 - Keep a stack of indices, not values.
 - The stack's values are kept in increasing order, from top to bottom.
 - So if you're finding the nearest number to the right that's bigger, the top of the stack is always the smallest unresolved value, and it gets bigger as you go down.

$\Rightarrow [3, 1, 4, 2]$ Array $[]$ Stack

$i=0$
 $val=3$ } Stack is empty. Nothing to compare.
Push index 0.

Stack $\rightarrow [0]$

Stack (as values) $\rightarrow [3]$

$\Rightarrow [3, 1, 4, 2]$

$i=1$
 $val=1$ } Compare to top of stack (value).
 $1 > 3$? No. 1 doesn't resolve anything. So we can push it.

Stack $\rightarrow [0, 1]$

Stack (as values) $\rightarrow [3, 1]$

$\Rightarrow [3, 1, 4, 2]$

$i=2$
 $val=4$ } Compare to top of stack (value).
 $4 > 1$? Yes. Index 1 has just found out its "nearest biggest to right", which is 4.

Distance $\rightarrow$
 $2 - 1 = 1$

Pop 1. Record answer. Discard it forever.

Now, stack has $[0]$, as values $[3]$.

Compare i with top of stack again. $4 > 3$?

Yes.

Distance $\rightarrow 2 - 0 = 2$

Pop 0. Record answer. Stack is now empty.

Push index 2.

Stack $\rightarrow [2]$

Stack (as values) $\rightarrow [4]$

$\Rightarrow [3, 1, 4, 2]$

$\begin{matrix} \uparrow \\ i=3 \\ \text{val}=2 \end{matrix} \left. \vphantom{\begin{matrix} \uparrow \\ i=3 \\ \text{val}=2 \end{matrix}} \right\} \text{Compare with top of stack.}$
 $2 > 4?$ No. Push it.

Stack $\rightarrow [2, 3]$

Stack (as values) $\rightarrow [4, 2]$

$\Rightarrow [3, 1, 4, 2]$ $\uparrow$ end reached. Anything left in the stack never found a bigger number. Their answer is none, so 0.

$\Rightarrow$ Final answers for $[3, 1, 4, 2]$

$\underbrace{2 \quad 1 \quad 5 \quad 0}_{\text{Answers}}$

Answers = $[2, 1, 0, 0]$

General Algorithm:

Stack = $[]$

for i in range(0, len(array)):

while stack is not empty and array

$[\text{stack.top}()] < \text{array}[i]$:

resolved_index = stack.pop()

Now we can do wtr we want
with the resolved index

stack.push(i)

Anything still in stack at the end never
found their answer.

For the comparison:

< 0 > → left-to-right / right-to-left
→ nearest bigger / nearest smaller